Topology-Aware Mapping for Photonic Interconnects in TPU-Class LLM Accelerators

Pranav Sitaraman • Simone Surdi
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1. Motivation

Google's TPU pods succeed because thousands of chips are connected by a carefully designed memory and interconnect system. TPU7x/Ironwood exposes 9216 chips per pod, 192 GiB HBM per chip, and 1200 GB/s bidirectional ICI per chip. For LLMs such as Gemma 4 and Gemini-class workloads, the hard part is often moving KV state, activations, expert tokens, gradients, and shared weights.

Question. If a future TPU pod used Lightmatter Passage-style photonic interconnect, would we only get lower link energy, or would the best distributed schedule change?

2. Underlying Principles

Class mapping split. We separate four layers: (1) einsum workload, (2) AccelForge architecture with `LogicalClusterMemory`, (3) inner AccelForge mapping (storage, temporal, spatial, compute), and (4) distributed schedule (XLA/GSPMD-style collectives, remote HBM, replicated cache, photonic multicast cache fill).

Network model.

$$E_{\text{net}} = B(e_c + h\,\ell(N)e_{tx}), \quad T_{\text{net}} = \frac{B}{BW/C(N)} + ht_r + h\,\ell(N)t_p + t_s.$$

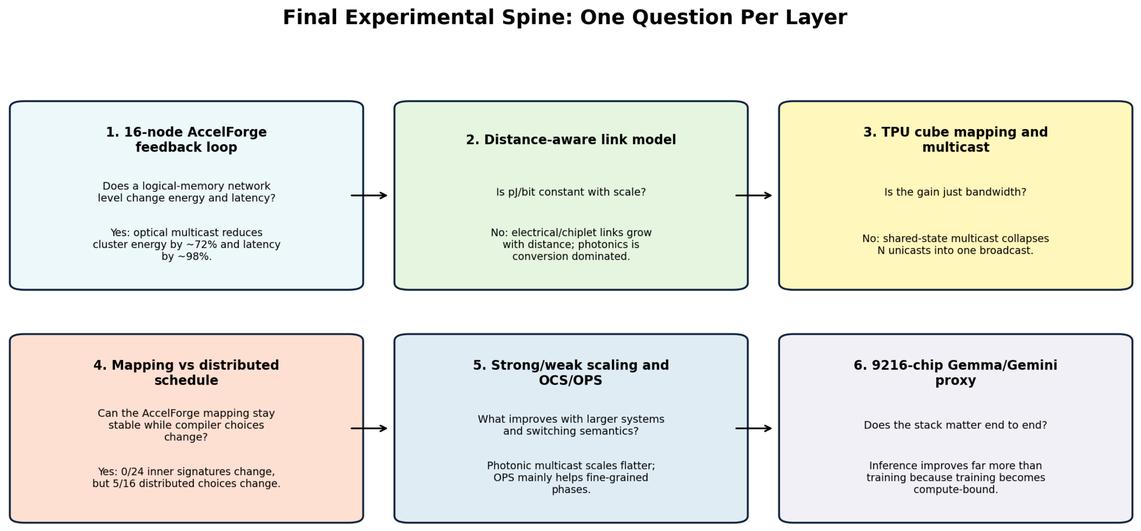
Electrical/chiplet links are distance sensitive; photonic links are conversion dominated. Non-multicast baselines pay repeated unicasts; photonic multicast pays the unique payload once.

AccelForge loop. mapper → logical-memory traffic → topology cost → updated logical-memory energy/latency → rerun until stable.

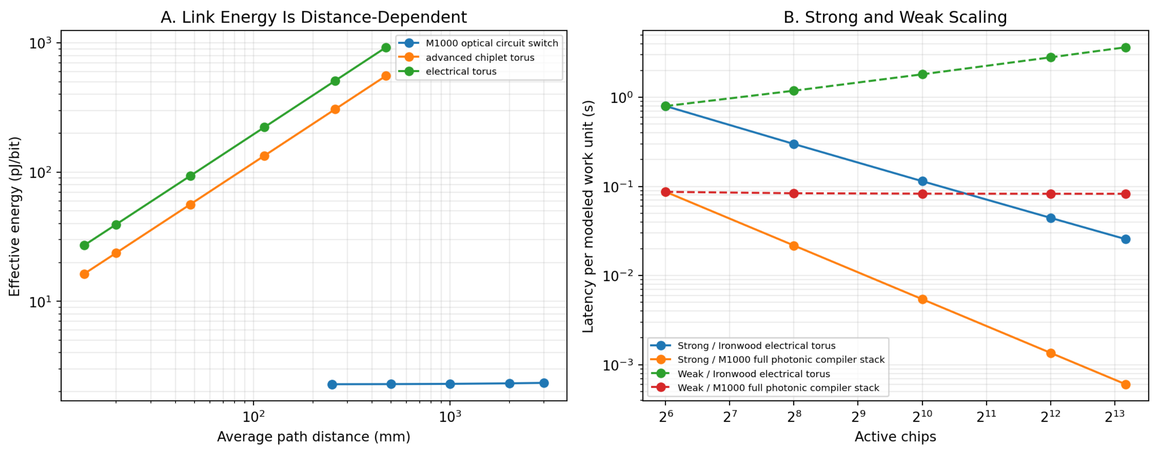
3. Evaluation Setup

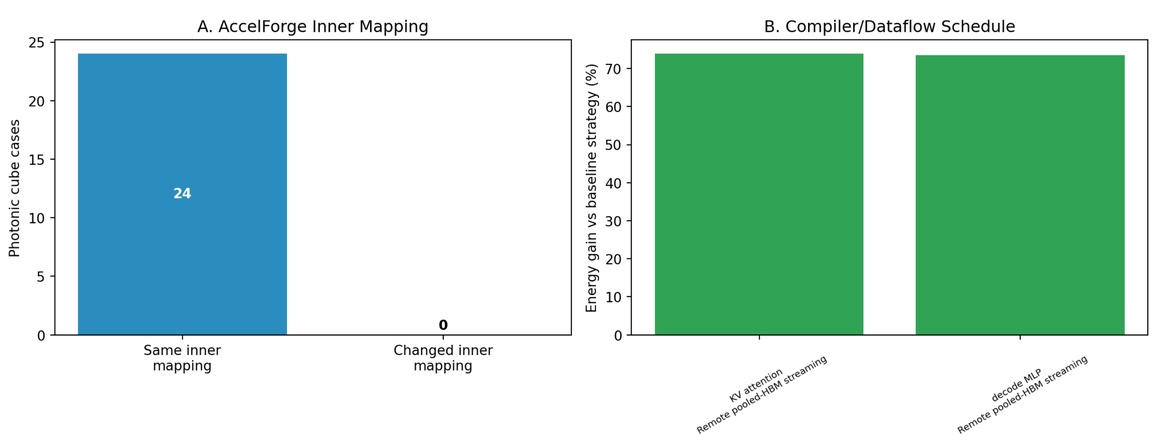
- **Fixed:** MAC/register/UB/HBM hierarchy and HBM cost.
- **Varied:** interconnect technology, topology, multicast, and distributed schedule.
- **Baselines:** electrical TPU torus and a strong 3.5 Tb/s, 6 pJ/bit chiplet interconnect.
- **Photonics:** EVK100 (3.2 Tb/s class, 1.9–3.2 pJ/bit) and M1000 (57.3 Tb/s one-way, 2.3 pJ/bit incl. laser, OCS).
- **Workloads:** six AccelForge GEMM proxies plus public Gemma 4 anchors and labelled Gemini-class stress proxies.

4. Final Experiment Spine

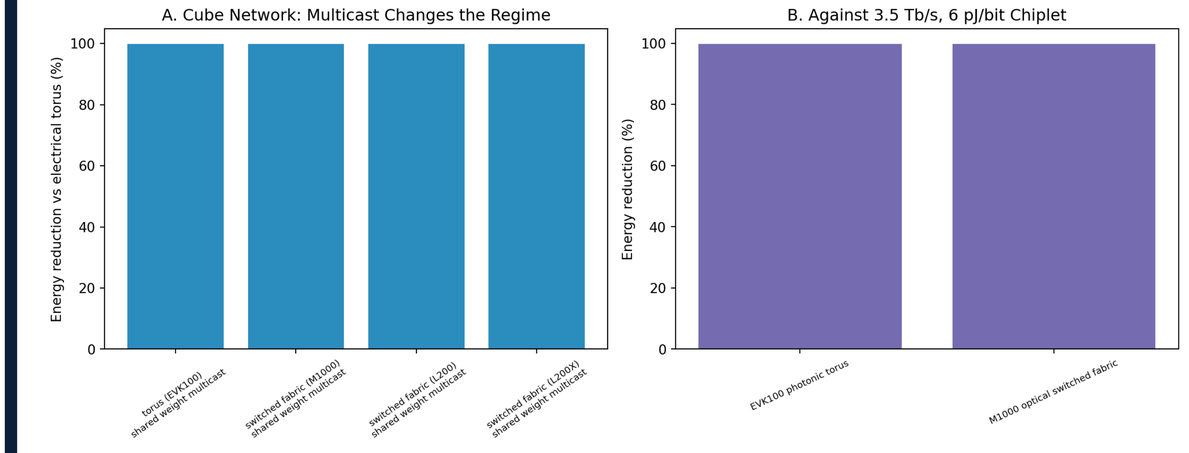


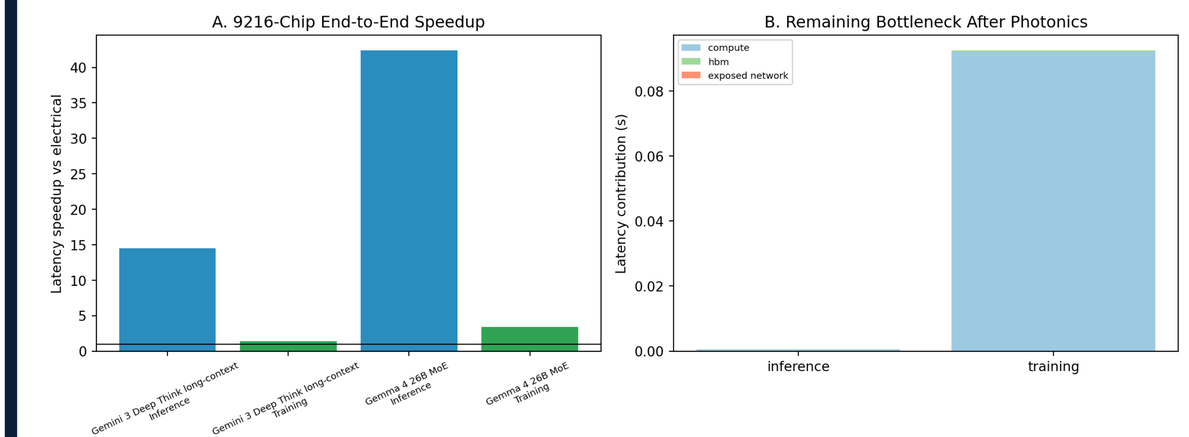
5. Results: Distance and Mapping





6. Results: Multicast and End-to-End





7. Analysis of Results

Finding	Evidence	Meaning
Logical network cost matters	16-node AccelForge: 71.95% energy and 97.57% latency reduction	Network-aware memory hierarchy changes system metrics.
Local mapping is stable	0/24 photonic TPU-cube inner signatures changed	Photonics does not magically alter local GEMM dataflow in this mapspace.
Distributed schedule changes	5/16 choices changed; best compiler-strategy gain 73.86%	Compiler-visible memory placement is where the new paradigm appears.
Multicast is decisive	99.93% TPU-cube network energy reduction; 99.85% vs chiplet on shared state	The key is one broadcast rather than N unicasts.
Inference benefits most	Gemma 4 26B MoE: 42.43× inference speedup, 3.37× training speedup	Decode/cache/expert routing exposes network stalls; large training becomes compute bound.

What we do not claim. The model does not include packet-level queues, exact production XLA code generation, area/cost, thermal drift, laser aging, or model-quality effects. These are the main future-work directions.

References and Reproducibility

Google TPU7x docs; Lightmatter Passage, EVK100, and M1000 EVK product pages; PyTorch/XLA SPMD guide; Gilbert *et al.*, "The Turbo-Charged Mapper," arXiv:2602.15172; 6.5930 lecture material on einsums, memory hierarchy, mapping, and advanced technologies.

Reproduce: `cd final-project && make study && make notebook && make test && make deliverables`. The final figures are in `workspace/final_project/results/presentation/`; raw CSVs remain in `workspace/final_project/results/`.

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